

Speak Like A Mathematician:

What does 7.3 ask, compared with 7.2?

Five Ways to Prove a Quadrilateral Is a Parallelogram

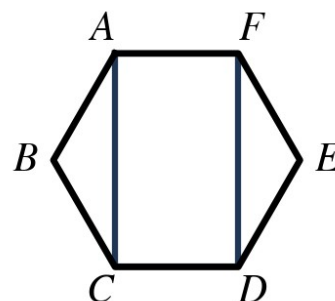
1. Both pairs of opposite sides are _____ (definition).
2. Both pairs of opposite sides are _____.
3. Both pairs of opposite angles are _____.
4. ONE pair of opposite sides is both _____ and _____.
5. The diagonals _____ each other.

Example 1 - Prove It with a Regular Hexagon

Given: ABCDEF is a regular hexagon.

Prove: ACDF is a parallelogram.

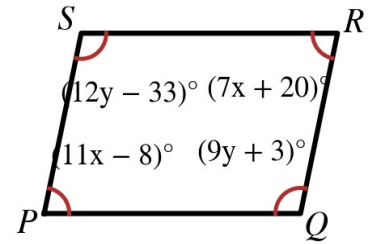
Statements	Reasons
1. ABCDEF is a regular hexagon	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.



Example 2 - Find x and y

In quadrilateral PQRS at the right, $m\angle P = (11x - 8)^\circ$, $m\angle R = (7x + 20)^\circ$, $m\angle Q = (9y + 3)^\circ$, and $m\angle S = (12y - 33)^\circ$. Find x and y so that PQRS is a parallelogram.

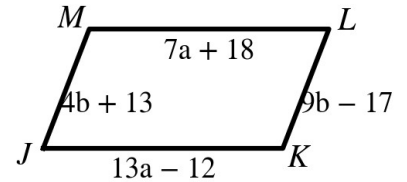
- Which condition applies? Solve for x.
- Solve for y, then check the angle sum.



Example 3 - Find a and b

In quadrilateral JKLM at the right, $JK = 13a - 12$, $LM = 7a + 18$, $KL = 9b - 17$, and $MJ = 4b + 13$. Find a and b so that JKLM is a parallelogram.

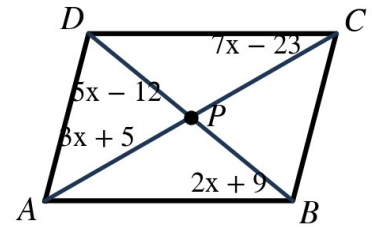
- Solve for a.
- Solve for b.



Example 4 - Diagonals That Bisect

In quadrilateral ABCD at the right, the diagonals meet at P. $AP = 3x + 5$, $PC = 7x - 23$, $BP = 2x + 9$, and $PD = 5x - 12$. Find x so that ABCD is a parallelogram.

- What must be true of the diagonals?
- Solve both equations. Do they agree?



Example 5 - A Coordinate Proof

Show that A(1, 4), B(6, 5), C(7, 1), D(2, 0) form a parallelogram, using the definition.

- Find the slopes of $\overline{AB}$ and $\overline{DC}$.
- Find the slopes of $\overline{BC}$ and $\overline{AD}$.
- State the conclusion.

